

Progressive block-size sweeps on TU-Berlin

Five families $\times$ two inits $\times$ four rates ($m = n = 8, 256 \times 256$)

Mean test PSNR (dB) over 50 sketches. n/a = stage undefined (mera needs k a power of 2).

1 High-resolution sketches (block-sparse)

TU-Berlin sketches are $\approx 96.5\%$ white — block-sparse — so block-DCT-8 is near-lossless at the light rate (≈ 100.7 dB mean @ $\rho = 0.20$) but collapses to ≈ 4.8 dB by $\rho = 0.01$.

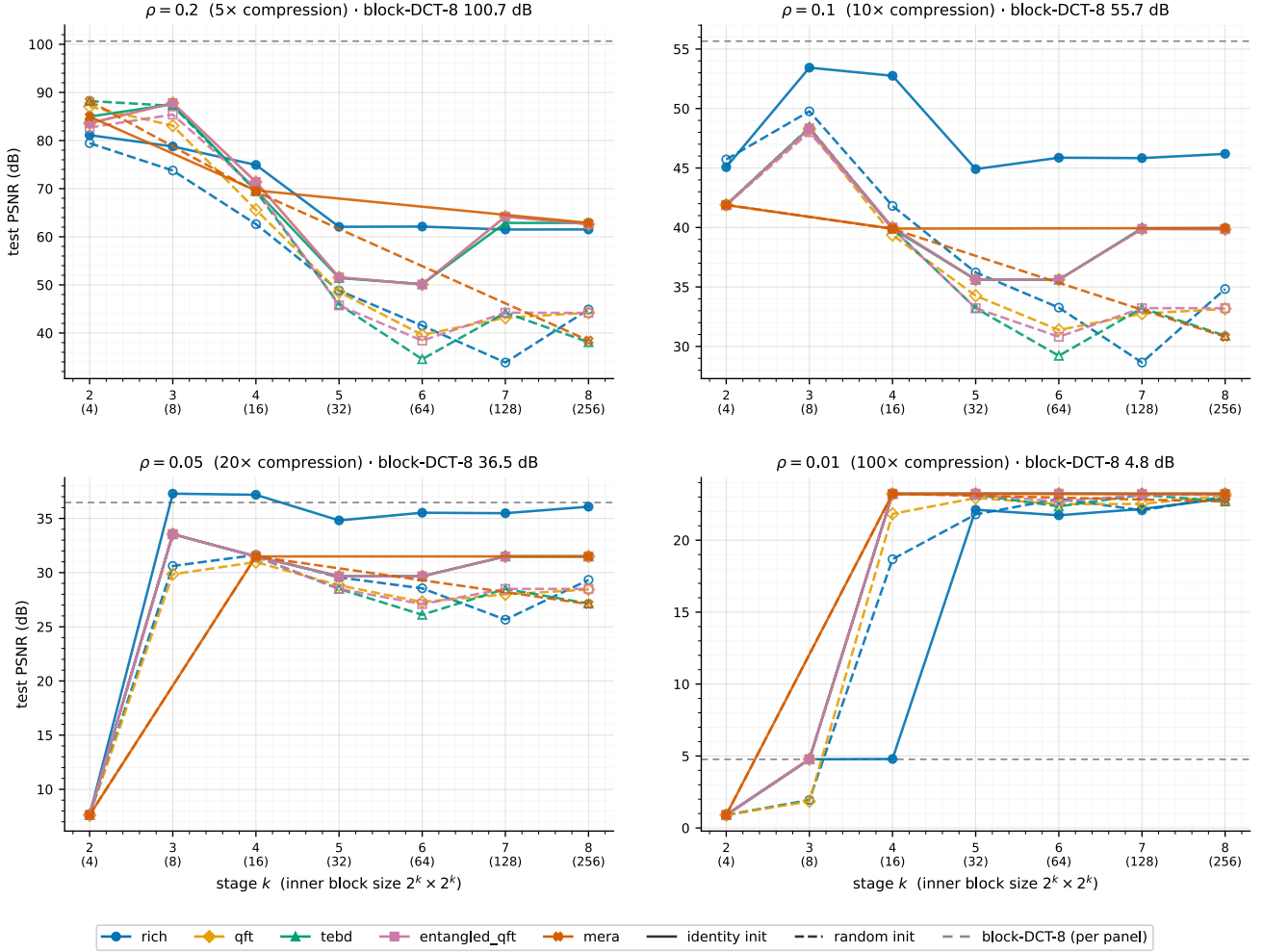


Figure 1: PSNR vs block size k at four keep ratios. One colour per family, solid = identity / dashed = random init; grey dashed = block-DCT-8 (value per panel).

2 Observations

- **The block-size curve inverts at the light rate** (small blocks win) — opposite of DIV2K.
- **Family ranking flips with rate**: QFT-family peaks ≈ 88 dB @ $\rho = 0.20$ ($k = 3$); rich leads @ $\rho = 0.05$; QFT-family again @ $\rho = 0.01$.
- **Heavy-compression crossover is extreme**: block-DCT $100.7 \rightarrow 4.8$ dB across the rates; learned circuits beat it by ≈ 18 dB at $\rho = 0.01$. Identity $\triangleright$ random (up to 25–30 dB).

3 Per-rate tables (mean PSNR, dB)

$\rho = 0.20 - 5\times$

family	init	$k=2 \setminus (4)$	$k=3 \setminus (8)$	$k=4 \setminus (16)$	$k=5 \setminus (32)$	$k=6 \setminus (64)$	$k=7 \setminus (128)$	$k=8 \setminus (256)$
rich	identity	81.1	78.8	74.9	62.1	62.1	61.5	61.5
rich	random	79.5	73.8	62.6	48.8	41.5	33.9	44.9

family	init	$k=2 \setminus (4)$	$k=3 \setminus (8)$	$k=4 \setminus (16)$	$k=5 \setminus (32)$	$k=6 \setminus (64)$	$k=7 \setminus (128)$	$k=8 \setminus (256)$
qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
qft	random	87.1	83.1	65.6	48.5	39.6	43.2	44.2
tebd	identity	85	87.5	69.6	51.4	50.2	62.9	62.9
tebd	random	88.2	87.2	69.5	45.9	34.6	44.2	38.1
ent-qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
ent-qft	random	82.7	85.4	70.1	45.7	38.4	44.2	44.2
mera	identity	85	n/a	69.6	n/a	n/a	n/a	62.9
mera	random	88.2	n/a	69.4	n/a	n/a	n/a	38.4

$\rho = 0.10 - 10\times$

family	init	$k=2 \setminus (4)$	$k=3 \setminus (8)$	$k=4 \setminus (16)$	$k=5 \setminus (32)$	$k=6 \setminus (64)$	$k=7 \setminus (128)$	$k=8 \setminus (256)$
rich	identity	45.1	53.4	52.7	44.9	45.9	45.8	46.2
rich	random	45.7	49.8	41.8	36.2	33.3	28.6	34.8
qft	identity	41.9	48.3	40	35.6	35.6	39.9	39.9
qft	random	41.9	48.2	39.4	34.3	31.4	32.8	33.2
tebd	identity	41.9	48.4	39.9	35.6	35.6	39.9	39.9
tebd	random	41.9	48.4	39.9	33.2	29.2	33.2	30.9
ent-qft	identity	41.9	48.3	40	35.6	35.6	39.9	39.9
ent-qft	random	41.9	48	39.9	33.2	30.8	33.2	33.2
mera	identity	41.9	n/a	39.9	n/a	n/a	n/a	39.9
mera	random	41.9	n/a	39.9	n/a	n/a	n/a	30.8

$\rho = 0.05 - 20\times$

family	init	$k=2 \setminus (4)$	$k=3 \setminus (8)$	$k=4 \setminus (16)$	$k=5 \setminus (32)$	$k=6 \setminus (64)$	$k=7 \setminus (128)$	$k=8 \setminus (256)$
rich	identity	7.6	37.3	37.2	34.8	35.5	35.5	36.1
rich	random	7.6	30.6	31.6	29.6	28.5	25.7	29.3
qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
qft	random	7.6	29.8	31	28.8	27.3	28	28.4
tebd	identity	7.6	33.5	31.5	29.7	29.7	31.5	31.5
tebd	random	7.6	33.5	31.5	28.5	26.1	28.5	27.1
ent-qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
ent-qft	random	7.6	33.5	31.5	28.5	27.1	28.5	28.5
mera	identity	7.6	n/a	31.5	n/a	n/a	n/a	31.5
mera	random	7.6	n/a	31.5	n/a	n/a	n/a	27.1

$\rho = 0.01 - 100\times$

family	init	$k=2 \setminus (4)$	$k=3 \setminus (8)$	$k=4 \setminus (16)$	$k=5 \setminus (32)$	$k=6 \setminus (64)$	$k=7 \setminus (128)$	$k=8 \setminus (256)$
rich	identity	0.9	4.8	4.8	22.1	21.7	22.2	22.9
rich	random	0.9	1.9	18.7	21.8	22.9	22.1	23
qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
qft	random	0.9	1.9	21.8	22.9	22.5	22.5	23.1
tebd	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
tebd	random	0.9	4.8	23.2	23.2	22.3	23.2	22.7
ent-qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
ent-qft	random	0.9	4.8	23.2	23.2	22.7	23.2	23.2
mera	identity	0.9	n/a	23.2	n/a	n/a	n/a	23.2
mera	random	0.9	n/a	23.2	n/a	n/a	n/a	22.7